Islam I. Abdulaal

Integrated Photonics Researcher

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RESEARCH INTERESTS

Integrated nanophotonics, quantum photonic sources, nonlinear on-chip devices, physics-informed inverse design, and wave-equation-constrained neural networks for photonic device optimization.

EDUCATION

M.Sc. in Advanced Optical Technologies (MAOT) Oct 2026 – Present
Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU) *Erlangen, Germany*

B.Sc. in Electronics and Communications Engineering Aug 2021 – Aug 2026
Alexandria University, Faculty of Engineering *Alexandria, Egypt*

RESEARCH EXPERIENCE

Research Intern Sep 2025 – Jul 2026
*NanoPhoto Lab, IMRE, A*STAR* *Remote*

Research Intern Jul 2025 – Sep 2025
Advanced Labs Program, Egypt Scholars Inc. *Remote*

Research Intern Jul 2023 – Aug 2025
OPST Group, SmartCI Research Center, Alexandria University *Hybrid*

SELECTED PUBLICATIONS

**Citation metrics based on Google Scholar (h-index: 3, citations: 23), October 2026.*

Journal Articles

[1] **Abdulaal, I. I.**, Elsayed, A. W. A., Abdelraouf, O. A. M., "Recent advances in terahertz-based bound states in the continuum metasurfaces for 6G communications and MedTech," *Journal of Optics*, 2026. DOI: 10.1088/2040-8986/ae8605 [Online]. Available: https://doi.org/10.1088/2040-8986/ae8605

Preprints

[2] **Abdulaal, I. I.**, Abdelraouf, O. A. M., "NanoPhotoNet-MPM: A physics-informed deep learning for inverse design of $NbOCl_2$ biphoton quantum sources," Preprints.org, 2026. DOI: 10.20944/preprints202609.1239.v1 [Online]. Available: https://doi.org/10.20944/preprints202609.1239.v1

[3] Labeb, A. H. M., Sallam, B., Elsayed, A. W., **Abdulaal, I. I.**, Kamal, A. M., Abdelraouf, O. A. M., "Beyond data-driven: How physics-informed neural networks are reshaping multi-physics design and discovery," arXiv:2606.21945, 2026. DOI: 10.48550/arXiv.2606.21945 [Online]. Available: https://arxiv.org/abs/2606.21945

HONORS, AWARDS, AND RESEARCH FUNDING

Max Planck School of Photonics (MPSP) Scholarship 2026 – Present
Issuer: Max Planck Schools

Dean's Honors for Academic Excellence 2021 – 2025
Issuer: Alexandria University, Faculty of Engineering

ICMTC Artificial Intelligence Contest (AIC-2) – 4th Place 2024
Issuer: Military Technical College

Huawei ICT Skills Competition – National Finalist (AI Track) 2023
Issuer: Huawei Technologies – MENA Region

NASA Space Apps Challenge – Global Nominee 2021
Issuer: NASA

LEADERSHIP EXPERIENCE AND ACADEMIC SERVICE

Board Member Jul 2026 – Present
Optica Egypt Local Section *Hybrid*

Chairman Feb 2025 – Oct 2025
IEEE Solid-State Circuits Society (SSCS), Alexandria University Student Branch Chapter *Hybrid*

General Coordinator Aug 2021 – Sep 2023
Education Clinic (USA-Registered NGO) *Remote*

PROFESSIONAL MEMBERSHIPS

- **Elite Network of Bavaria** Oct 2026 – Present
  Student Member.
- **Optica (formerly OSA)** Jul 2026 – Present
  Student Member.
- **Institute of Electrical and Electronics Engineers (IEEE)** 2025 – 2026
  Student Member (ID: 101099759).

REFERENCES

Dr. Omar Abdelraouf
Research Scientist
Institute of Materials Research and Engineering (IMRE), A*STAR, Singapore
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Prof. Ishac Kandas
Professor
Department of Engineering Physics and Mathematics, Alexandria University
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